# Applied Econometrics: Session 7

## Difference-in-Differences, Panel Data, and Fixed Effects

PhD Intensive Course

Lecture 7

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# How This Lecture Fits the Course

Earlier lectures asked how to make treated and untreated units comparable.

Today we use a different source of comparison:

**Core idea**

If treated and untreated units would have followed the same trend without treatment, then the control group's change can proxy for the treated group's missing counterfactual change.

- Difference-in-differences uses comparisons across groups and across time.
- Panel fixed effects use repeated observations to remove fixed unit differences.
- Both methods are powerful, but only under clear trend assumptions.

# Roadmap for Today

1. The 2-by-2 DiD Design
2. Event-Study Intuition
3. Panel Data
4. Practical Diagnostics
5. Synthesis

# Motivating Example: A City Policy

Suppose one city introduces a new job-search program in 2025. We observe average employment in:

- the treated city before and after the program;
- a similar control city before and after the program.

**Causal question**

What would employment in the treated city have been after 2025 if the program had not happened?

**Why this is hard**

We observe the treated city after treatment, but not the same treated city after no treatment.

# The Four Cells

Let $D_i = 1$ denote the treated group and $Post_t = 1$ denote the period after treatment begins.

| | $Post_t = 0$ | $Post_t = 1$ |
|---|---|---|
| $D_i = 0$ control | $\bar{Y}_{00}$ | $\bar{Y}_{01}$ |
| $D_i = 1$ treated | $\bar{Y}_{10}$ | $\bar{Y}_{11}$ |

- $\bar{Y}_{10}$: treated group before treatment.
- $\bar{Y}_{11}$: treated group after treatment.
- $\bar{Y}_{00}, \bar{Y}_{01}$: control group before and after.

# Two Naive Comparisons

**Before-after for treated:**

$$\bar{Y}_{11} - \bar{Y}_{10}.$$

This is biased if the outcome would have changed over time even without treatment.

**Post-period treated-control gap:**

$$\bar{Y}_{11} - \bar{Y}_{01}.$$

This is biased if treated and control groups differ in baseline levels.

**DiD's move**

Use the control group's change over time to remove common trends, and use the treated group's pre-period level to remove fixed baseline differences.

# The Difference-in-Differences Estimator

The 2-by-2 DiD estimator is

$$\hat{\delta}_{\mathrm{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}).$$

Equivalent form:

$$\hat{\delta}_{\mathrm{DiD}} = (\bar{Y}_{11} - \bar{Y}_{01}) - (\bar{Y}_{10} - \bar{Y}_{00}).$$

- First expression: difference in time changes.
- Second expression: difference in group gaps.
- Same number, two useful interpretations.

Appendix details

# Numeric Example

| | Before | After |
|---|---|---|
| Control city | 50 | 58 |
| Treated city | 40 | 55 |

$$\text{Treated change} = 55 - 40 = 15, \qquad \text{Control change} = 58 - 50 = 8.$$

$$\hat{\delta}_{\mathrm{DiD}} = 15 - 8 = 7.$$

**Interpretation**

Employment rose by 15 in the treated city, but 8 of those points look like a common trend. The remaining 7 is the DiD estimate.

# Counterfactual Picture in Words

DiD constructs the missing treated-after-without-treatment outcome:

$$\mathbb{E}[Y_0 \mid \widehat{D = 1, Post} = 1] = \bar{Y}_{10} + (\bar{Y}_{01} - \bar{Y}_{00}).$$

In the numeric example:

$$\widehat{Y}_{\text{treated, after, no treatment}} = 40 + (58 - 50) = 48.$$

We actually observe 55, so the estimated effect is

$$55 - 48 = 7.$$

**Student translation**

Take the treated group's starting point. Add the control group's growth. Compare the observed treated-after outcome to that counterfactual.

# Potential Outcomes Notation

Let $Y_{it}(1)$ be unit $i$'s outcome at time $t$ if treated, and $Y_{it}(0)$ if untreated. Treatment starts only in the post period for the treated group.

$$Y_{it} = Y_{it}(0) + D_i Post_t \left( Y_{it}(1) - Y_{it}(0) \right).$$

The target is the average treatment effect on the treated after treatment:

$$\mathrm{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1].$$

The missing object is

$$\mathbb{E}[Y_{i1}(0) \mid D_i = 1].$$

Appendix details

# Parallel Trends

Parallel trends says that, absent treatment, treated and control groups would have had the same average change:

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0].$$

This does **not** require:

- equal levels before treatment;
- equal covariate distributions;
- random assignment of treatment.

It requires equal untreated **trends**.

Appendix details

# What Parallel Trends Allows

Parallel trends allows fixed differences in levels:

$$\mathbb{E}[Y_{i0}(0) \mid D = 1] \neq \mathbb{E}[Y_{i0}(0) \mid D = 0].$$

Example:

- Rich cities can have higher employment than poor cities.
- That is not fatal if both would have grown by the same amount without treatment.
- DiD subtracts out the level difference by comparing changes.

**Important distinction**

A control group can be useful even if it starts above or below the treated group. It becomes dangerous when it is on a different path.

# What Breaks Parallel Trends

Parallel trends fails when treatment timing is related to untreated growth. Common examples:

- A city receives a job program because unemployment was already worsening.
- A firm advertises in markets where sales were already falling.
- A school adopts a curriculum because test scores had begun rising.
- A policy is timed to a local shock that affects the outcome.

**Diagnostic question**

Would the treated group have changed like the control group if the treatment had not happened?

# Regression Version of 2-by-2 DiD

Estimate

$$Y_{it} = \alpha + \gamma D_i + \lambda Post_t + \delta(D_i \times Post_t) + u_{it}.$$

The coefficient $\delta$ equals the 2-by-2 DiD estimator.

$$\hat{\delta} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}).$$

- $D_i$: permanent treated-control level difference.
- $Post_t$: common post-period shock.
- $D_i \times Post_t$: treatment turns on for treated units after policy.

Appendix details

# Why Regression Is Convenient

The regression form gives the same point estimate in the simple 2-by-2 design, but adds:

- standard errors;
- covariates, if pre-treatment and justified;
- many periods and many units;
- fixed effects;
- event-study coefficients.

**Warning**

Regression does not make parallel trends true. It only gives a convenient way to estimate the design.

# Why More Pre-Periods Help

With only one pre-period, parallel trends is not visually testable. With several pre-periods, we can ask:

> Were treated and control groups already moving differently before treatment?

This is not a proof of parallel trends, because the unobserved post-treatment counterfactual is still missing. But it is an important credibility check.

**Practical rule**

If treated and control groups diverge before treatment, be cautious about interpreting the post-treatment divergence as causal.

# Event Time

Event studies re-index time relative to treatment:

$$k = t - T_i^*,$$

where $T_i^*$ is the treatment start date.

- $k = -3$: three periods before treatment.
- $k = -1$: the omitted reference period, usually just before treatment.
- $k = 0$: first treated period.
- $k = 2$: two periods after treatment.

**Goal**

Estimate how the treated-control difference evolves before and after the treatment date.

# A Simple Event-Study Regression

With one treated cohort and never-treated controls:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \left(1\{t - T_i^* = k\} \times D_i\right) + u_{it}.$$

- Unit fixed effects absorb permanent treated-control differences.
- Time fixed effects absorb common shocks.
- $\delta_k$ compares treated units to controls at event time $k$, relative to $k = -1$.

Appendix details

# How to Read Event-Study Coefficients

- Pre-treatment coefficients ($k < 0$) are called leads.
- Post-treatment coefficients ($k \geq 0$) are called lags or dynamic effects.
- Leads near zero support the idea that groups were not already diverging.
- A jump at $k = 0$ suggests an immediate treatment effect.
- A gradual increase after $k = 0$ suggests delayed adjustment.

**Student warning**

Do not say "the pre-trend test proves parallel trends." It can only fail to detect obvious pre-treatment violations.

# Event-Study Pitfalls

- Few pre-periods mean weak diagnostics.
- Wide confidence intervals can hide economically important pre-trends.
- Anticipation effects can move outcomes before treatment begins.
- Changing composition can make the treated group different over time.
- With staggered adoption and heterogeneous effects, simple TWFE event studies can be misleading.

**For this course**

Today we focus on the clean case: one treated cohort, never-treated controls, and a common treatment start date.

# What Is Panel Data?

Panel data repeatedly observes the same unit over time:

$$i = 1, \ldots, N, \qquad t = 1, \ldots, T.$$

Examples:

- people observed in multiple survey waves;
- firms observed across years;
- municipalities observed monthly;
- schools observed before and after a reform.

**Why panels matter**

Repeated observations let us compare a unit to itself over time, not only to other units.

# Panel Notation

$$Y_{it} = \text{outcome for unit } i \text{ in period } t.$$

$$D_{it} = \text{treatment status for unit } i \text{ in period } t.$$

$$X_{it} = \text{observed covariates for unit } i \text{ in period } t.$$

Panel regressions must respect two dimensions:

- observations from the same unit are not independent over time;
- shocks in the same period can affect many units at once.

# Why Panels Help with Unobserved Confounding

Suppose ability, geography, management quality, or culture affects the outcome. If that feature is fixed for unit $i$, write it as $a_i$.

$$Y_{it} = \beta D_{it} + a_i + u_{it}.$$

- Cross-sectional OLS confuses $D_{it}$ with $a_i$ if treated units are permanently different.
- Fixed effects compare each unit to its own average.
- Anything constant within the unit is removed.

**Plain English**

Fixed effects control for stable differences that we may not observe but that travel with the unit over time.

# The Unit Fixed-Effects Model

A unit fixed-effects model is

$$Y_{it} = \beta D_{it} + a_i + u_{it}.$$

$a_i$ is a unit-specific intercept.

- Each person, firm, school, or city gets its own baseline.
- Identification comes from within-unit changes in treatment.
- Units whose treatment never changes do not identify $\beta$, though they may help estimate time effects in richer models.

Appendix details

# The Within Transformation

For each unit define time averages:

$$\bar{Y}_i = \frac{1}{T_i}\sum_t Y_{it}, \qquad \bar{D}_i = \frac{1}{T_i}\sum_t D_{it}.$$

Subtract the unit average:

$$Y_{it} - \bar{Y}_i = \beta(D_{it} - \bar{D}_i) + (u_{it} - \bar{u}_i).$$

The fixed unit term disappears because $a_i - \bar{a}_i = 0$.

**Mechanical meaning**

FE regression is OLS on demeaned outcomes and demeaned treatment.

Appendix details

# What Fixed Effects Cannot Estimate

In a unit FE model, time-invariant variables are absorbed:

$$Z_i - \bar{Z}_i = 0.$$

Examples:

- sex, birth cohort, ethnicity, country of birth;
- distance to the coast, historical institutions, baseline geography;
- any variable measured once and copied across all years.

**Interpretation**

FE can control for time-invariant variables, but cannot estimate their separate coefficients in the same model.

# Time Fixed Effects

A time fixed effect is a period-specific intercept:

$$Y_{it} = \beta D_{it} + a_i + \lambda_t + u_{it}.$$

$\lambda_t$ absorbs shocks common to all units in period $t$:

- national recessions;
- inflation;
- pandemics;
- common seasonality;
- changes in national measurement rules.

**Two-way fixed effects**

Unit FE compare units to themselves; time FE remove shocks shared by everyone at a point in time.

# Two-Way Fixed Effects and DiD

The basic TWFE DiD regression is

$$Y_{it} = a_i + \lambda_t + \delta D_{it} + u_{it}.$$

In a simple 2-by-2 design:

$$D_{it} = 1\{i \text{ treated}\} \times 1\{t \text{ post}\}.$$

Then $\hat{\delta}$ equals the 2-by-2 DiD estimate.

- $a_i$ replaces the treated-group dummy.
- $\lambda_t$ replaces the post dummy.
- $D_{it}$ is the interaction that turns treatment on.

Appendix details

# Simple TWFE Equivalence

Compare the two regressions:

$$Y_{it} = \alpha + \gamma D_i + \lambda Post_t + \delta(D_i Post_t) + u_{it}$$

and

$$Y_{it} = a_i + \lambda_t + \delta D_{it} + u_{it}.$$

In the 2-by-2 case, $D_{it} = D_i Post_t$.

**Same estimator, different notation**

The first expression is easier for beginners. The second scales to many units and many periods.

# Fixed-Effect Demeaning with Two-Way FE

For balanced panels, two-way demeaning removes unit and time averages:

$$\tilde{Y}_{it} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{Y}.$$

$$\tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}.$$

Then TWFE estimates $\delta$ by regressing $\tilde{Y}_{it}$ on $\tilde{D}_{it}$.

**Intuition**

TWFE uses only the outcome variation left after removing each unit's average and each time period's average.

# Checklist Before Running DiD

1. Define the treatment date and treatment group precisely.
2. Plot raw means for treated and control groups over time.
3. Check whether pre-treatment trends look similar.
4. Explain why treatment timing was not driven by unobserved outcome trends.
5. Decide the level at which treatment varies and cluster standard errors there.

**Design first**

A table of coefficients is not a substitute for a credible comparison group.

# What to Plot

Useful figures:

- treated and control mean outcomes over time;
- the treated-control gap over time;
- event-study coefficients with confidence intervals;
- number of units observed by group and period;
- distribution of treatment timing if adoption is staggered.

**What you are looking for**

Is there a visible break at treatment time, or were groups already moving apart before treatment?

# Standard Errors

Repeated observations create dependence:

$$u_{it} \text{ and } u_{is} \quad \text{may be correlated for the same unit } i.$$

Practical guidance:

- cluster at the treatment-assignment level when possible;
- if policy varies by state, cluster by state;
- if policy varies by school, cluster by school;
- with very few clusters, conventional cluster-robust standard errors can be unreliable.

**Core point**

DiD standard errors must respect the level at which shocks and treatment assignment are correlated.

# Covariates in DiD

Covariates can help precision or adjust for composition, but they do not rescue a bad design. Safer covariates:

- pre-treatment characteristics;
- time-varying controls that are clearly not affected by treatment.

Risky covariates:

- post-treatment outcomes or mediators;
- variables that treatment itself changes;
- controls chosen after looking at the desired coefficient.

# When Panel Fixed Effects Do Not Solve the Problem

Fixed effects remove only time-invariant unobserved confounding. They do not remove:

- unobserved shocks that change over time within units;
- reverse causality;
- anticipation before treatment;
- spillovers from treated to control units;
- measurement changes that differ by group;
- differential trends not captured by the model.

**Honest conclusion**

Panel data is useful because it changes the identifying assumption. It does not eliminate the need for an identifying assumption.

# Common Reporting Template

A clear DiD write-up should state:

- outcome, treatment, treated group, control group, and time window;
- the exact regression specification;
- the identifying assumption in words;
- evidence on pre-trends;
- standard-error clustering level;
- robustness checks and limitations.

**One-sentence identification claim**

"Absent the policy, treated and control units would have experienced the same average change in the outcome over the study window."

# Python Example for This Lecture

The companion script is:

`lectures/code/lecture-7-did-panel.py`

It simulates:

- a panel where parallel trends holds;
- a panel where treated units have a different untreated trend;
- 2-by-2 DiD estimates;
- TWFE estimates;
- event-study coefficients.

**What students should notice**

The estimator works in the first simulation and is biased in the second even though the regression runs normally in both.

# Main Takeaways

1. DiD estimates treatment effects by comparing changes, not levels.
2. The key identifying assumption is parallel untreated trends.
3. Event studies make the trend assumption visible, but do not prove it.
4. Unit fixed effects remove stable unit differences.
5. Time fixed effects remove common period shocks.
6. In the clean 2-by-2 case, TWFE and the DiD formula give the same estimate.

# Source and Further Reading

This lecture is based on ideas from:

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 13, "Difference-in-Differences".
- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 14, "Panel Data and Fixed Effects".
- Angrist and Pischke, *Mostly Harmless Econometrics*, chapters on panel data and differences-in-differences.
- Cunningham, *Causal Inference: The Mixtape*, chapters on DiD and fixed effects.

Handbook links:

- `https://matheusfacure.github.io/python-causality-handbook/13-Difference-in-Differences.html`
- `https://matheusfacure.github.io/python-causality-handbook/14-Panel-Data-and-Fixed-Effects.html`

# What Students Should Be Able to Do

After this lecture, students should be able to:

- compute a 2-by-2 DiD estimate by hand;
- state the parallel trends assumption in words and symbols;
- explain the purpose of event-study lead coefficients;
- write a panel regression with unit and time fixed effects;
- explain why demeaning removes unit fixed effects;
- describe practical diagnostics before trusting a DiD estimate.

# Appendix Map

- A1: 2-by-2 DiD algebra.
- A2: DiD identification under parallel trends.
- A3: Regression coefficient equals DiD.
- A4: Event-study coefficients.
- A5: Within transformation for one-way fixed effects.
- A6: TWFE equivalence in the 2-by-2 design.

# Appendix A1: 2-by-2 DiD Algebra

Start from the change comparison:

$$(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}).$$

Rearrange terms:

$$\bar{Y}_{11} - \bar{Y}_{10} - \bar{Y}_{01} + \bar{Y}_{00} = \bar{Y}_{11} - \bar{Y}_{01} - \bar{Y}_{10} + \bar{Y}_{00}.$$

Group as post gap minus pre gap:

$$(\bar{Y}_{11} - \bar{Y}_{01}) - (\bar{Y}_{10} - \bar{Y}_{00}).$$

The two formulas are numerically identical.

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# Appendix A2: DiD Identification Under Parallel Trends

The target is

$$\mathrm{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1].$$

The observed treated change is

$$\begin{aligned}\mathbb{E}[Y_{i1} \mid D = 1] - \mathbb{E}[Y_{i0} \mid D = 1] &= \mathbb{E}[Y_{i1}(1) \mid D = 1] - \mathbb{E}[Y_{i0}(0) \mid D = 1] \\ &= \mathrm{ATT} + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D = 1].\end{aligned}$$

The treated change equals the causal effect plus the untreated trend for treated units.

# Appendix A2 Continued: Subtract the Control Trend

Under parallel trends,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D = 0].$$

Since controls are untreated in both periods,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D = 0] = \mathbb{E}[Y_{i1} - Y_{i0} \mid D = 0].$$

Therefore

$$\mathrm{ATT} = \{\mathbb{E}[Y_{i1} \mid D = 1] - \mathbb{E}[Y_{i0} \mid D = 1]\} - \{\mathbb{E}[Y_{i1} \mid D = 0] - \mathbb{E}[Y_{i0} \mid D = 0]\}.$$

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# Appendix A3: Regression Coefficient Equals DiD

In the saturated 2-by-2 regression

$$Y_{it} = \alpha + \gamma D_i + \lambda Post_t + \delta(D_i Post_t) + u_{it},$$

predicted cell means are:

| | $Post = 0$ | $Post = 1$ |
|---|---|---|
| $D = 0$ | $\alpha$ | $\alpha + \lambda$ |
| $D = 1$ | $\alpha + \gamma$ | $\alpha + \gamma + \lambda + \delta$ |

The treated change is $\lambda + \delta$. The control change is $\lambda$. Difference:

$$(\lambda + \delta) - \lambda = \delta.$$

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# Appendix A4: Event-Study Coefficients

Let $k = t - T_i^*$ be event time and omit $k = -1$.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k 1\{k = t - T_i^*\} D_i + u_{it}.$$

The omitted period $k = -1$ is the reference. Each $\delta_k$ is interpreted relative to that period:

$$\delta_k = (\text{treated-control gap at event time } k) - (\text{treated-control gap at } k = -1),$$

after removing unit and calendar-time fixed effects.

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# Appendix A5: Within Transformation

Start with

$$Y_{it} = \beta D_{it} + a_i + u_{it}.$$

Average over time within unit $i$:

$$\bar{Y}_i = \beta \bar{D}_i + a_i + \bar{u}_i.$$

Subtract:

$$\begin{aligned} Y_{it} - \bar{Y}_i &= \beta(D_{it} - \bar{D}_i) + (a_i - a_i) + (u_{it} - \bar{u}_i) \\ &= \beta(D_{it} - \bar{D}_i) + (u_{it} - \bar{u}_i). \end{aligned}$$

The unit fixed effect $a_i$ is gone.

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# Appendix A6: TWFE Equivalence in the 2-by-2 Design

With two groups and two periods, write

$$D_{it} = G_i Post_t,$$

where $G_i = 1$ for treated units. The TWFE model is

$$Y_{it} = a_i + \lambda_t + \delta D_{it} + u_{it}.$$

If we replace unit fixed effects by a group dummy and time fixed effects by a post dummy, this becomes

$$Y_{it} = \alpha + \gamma G_i + \lambda Post_t + \delta(G_i Post_t) + u_{it}.$$

By Appendix A3, the coefficient on $G_i Post_t$ is exactly the DiD estimator. Back to main slide